Homework 1 - Analysis of Brazilian Assets

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1 Introduction

We chose to analyze 15 Brazilian assets, taken from the Bovespa index, with different market capitalizations and different sectors.

Asset	**Ticker**	**Mkt Cap (USD bn)**	**Sector / Industry**
Petrobras	PETR4.SA	136.4	Energy / Oil & Gas Integrated
Itaú Unibanco	ITUB4.SA	79.0	Financial Services / Banks
Vale	VALE3.SA	48.3	Basic Materials / Metals & Mining
Ambev	ABEV3.SA	41.5	Consumer Defensive / Brewers
WEG	WEGE3.SA	40.0	Industrials / Electrical Equipment
Banco Bradesco	BBDC4.SA	28.6	Financial Services / Banks
Banco do Brasil	BBAS3.SA	18.7	Financial Services / Banks
Telefônica Brasil	VIVT3.SA	17.9	Communication Services / Telecom
Sabesp	SBSP3.SA	15.9	Utilities / Regulated Water
Embraer	EMBJ3.SA	13.4	Industrials / Aerospace & Defense
Eneva	ENEV3.SA	9.5	Utilities / Diversified
Gerdau	GGBR4.SA	9.1	Basic Materials / Steel
Equatorial	EQTL3.SA	8.1	Utilities / Regulated Electric
Copel	CPLE3.SA	7.7	Utilities / Regulated Electric
Localiza	RENT3.SA	6.2	Industrials / Rental & Leasing

Table 1: Selected assets. Market capitalization of November 2026.

We concentrated more on Energy and utility companies, as they are more sensitive to the Brazilian macroeconomic environment, and due to the recent energy crisis that characterised these years. We also included some banks and industrials to diversify the sample.

2 Q1 - Data

Reference: DATA folder, `01_clean_data.py`, `logreturns_brazil.csv`, `prices_brazil_clean.csv`.

3 Results

Numerical analysis and outcomes.

4 Conclusion

Final remarks.